Chapter 7

Linear Evolution Equations

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 7; printed p. 349; PDF page 366.

Chinese reference: 中译本对应章：线性发展方程，书内第 248 页起。

Study edition / 英文重构学习稿。 The exposition follows Evans's section order. Source theorem identifiers are section-local; the displayed theorem numbers belong to these notes. All proofs, Chinese mathematical annotations, exercise solutions, and TikZ diagrams are editorial reconstructions or supplements, not verbatim transcriptions. The chapter has not received independent mathematical review.

We pass from elliptic boundary problems to trajectories in function spaces. Parabolic equations dissipate energy; hyperbolic equations transport it. Galerkin approximation treats time-dependent coefficients, Fourier analysis detects matrix structure, and semigroups organize autonomous evolution through elliptic resolvents. Constants may depend on the domain, final time, ellipticity, and stated coefficient norms, never on the approximation index.

7.1 Second-Order Parabolic Equations

7.1.1 The evolution space and the initial trace

Let $U \subset \mathbb{R}^n$ be bounded with smooth boundary, $T > 0$, and set $Q_T = U \times (0,T)$, $V = H_0^1(U)$, $H = L^2(U)$, $V' = H^{-1}(U)$. Identify $H \subset V'$ by $\langle h, v\rangle = (h,v)_H$. Consider

$$u_t + L(t)u = f, \qquad u|_{\partial U} = 0, \qquad u(0) = g, \tag{7.1}$$

where repeated spatial indices are summed and

$$L(t)u = -\partial_j(a^{ij}(x,t)\partial_i u) + b^i(x,t)\partial_i u + c(x,t)u.$$

Initially assume real bounded measurable coefficients, $a^{ij} = a^{ji}$, and $a^{ij}\xi_i\xi_j \geq \theta|\xi|^2$ almost everywhere, uniformly in time. The bilinear form

$B_t[u,v] = \int_U (a^{ij}u_i v_j + b^i u_i v + cuv)\,dx$ satisfies

$$\begin{aligned} |B_t[u,v]| &\le M\|u\|_V\|v\|_V, \\ B_t[v,v] &\ge \alpha\|v\|_V^2 - \gamma\|v\|_H^2 \end{aligned} \tag{7.2}$$

with $M, \alpha > 0$, $\gamma \ge 0$, uniform in t. Indeed $|\int b^i u_i u| \le \varepsilon\|Du\|_2^2 + C_\varepsilon\|u\|_2^2$; a negative potential is absorbed into γ.

时间导数的对偶空间

散度项把 H_0^1 函数送入 H^{-1}。弱解通常只有 $u_t \in L^2(0,T;H^{-1})$，不能先把 u_t 当成 $L^2(U)$ 函数。

Lemma 7.1 (Time trace and energy identity; cf. source §5.9.2, Theorem 3)
If $u \in L^2(0,T;V)$ and $u' \in L^2(0,T;V')$, then u has a unique representative in $C([0,T];H)$, and

$$\|u(t)\|_H^2 - \|u(s)\|_H^2 = 2\int_s^t \langle u', u\rangle\,dr. \tag{7.3}$$

The analogous product rule holds for two such functions.

Expanded proof (editorial reminder of time regularization).

For smooth V-valued curves this is differentiation of an inner product. Such curves are dense in $\|u\|_{L^2V} + \|u'\|_{L^2V'}$ by the time approximation theorem in §5.9: extend by reflection at the endpoints in V', then mollify in time. For a smooth curve z, choose s with $\|z(s)\|_H^2 \le T^{-1}\|z\|_{L^2H}^2$ and integrate its derivative:

$$\sup_t \|z(t)\|_H^2 \le T^{-1}\|z\|_{L^2H}^2 + 2\|z'\|_{L^2V'}\|z\|_{L^2V}.$$

Applied to differences, this gives uniform convergence in H. The products converge in L^1, so the identity passes to every pair of endpoints. Polarization gives the product rule. Equality almost everywhere makes the continuous representative unique. □

Definition 7.2 (Weak parabolic solution; source §7.1.1)
For $f \in L^2(0,T;V')$, $g \in H$, a weak solution belongs to $L^2(0,T;V)$, has $u' \in L^2(0,T;V')$, satisfies $u(0) = g$ in H, and obeys

$$\langle u'(t), v\rangle + B_t[u(t), v] = \langle f(t), v\rangle \quad (v \in V) \quad \text{for a.e. } t.$$

Allowing V'-valued forcing is an editorial extension of the source's $L^2(Q_T)$ forcing. Regularity statements below specify stronger data.

7.1.2 Galerkin construction, estimates, and uniqueness

Let w_k be the H-orthonormal Dirichlet eigenfunctions of $-\Delta$. They are orthogonal for $(u,v)_V = \int_U (Du\cdot Dv + uv)$. The H-projection P_m onto $V_m = \operatorname{span}(w_1,\dots,w_m)$ satisfies $\|P_m v\|_V \le \|v\|_V$ and $P_m v \to v$ in V. Seek $u_m = \sum_{k\le m} d_k(t) w_k$ with

$$\begin{aligned} d_k' + \sum_{\ell\le m} B_t[w_\ell, w_k] d_\ell &= \langle f, w_k\rangle, \\ d_k(0) &= (g, w_k). \end{aligned} \tag{7.4}$$

This linear system with measurable bounded coefficients has a unique absolutely continuous solution: solve the integral equation by contraction on short intervals where $\int \|E(t)\|dt < 1$, then concatenate them.

Theorem 7.3 (Weak well-posedness; source §7.1.2, Theorems 1–4)
There is exactly one weak solution, and

$$\sup_{0\le t\le T} \|u(t)\|_H^2 + \|u\|_{L^2V}^2 + \|u'\|_{L^2V'}^2 \le C\big(\|g\|_H^2 + \|f\|_{L^2V'}^2\big). \tag{7.5}$$

The same estimate holds for differences of solutions and their data.

Expanded proof (uniform bounds and recovery of the initial value).

Multiply the finite equations by d_k and sum. Using (7.2) and Young's inequality gives

$$\frac{1}{2}\frac{d}{dt}\|u_m\|_H^2 + \frac{\alpha}{2}\|u_m\|_V^2 \le \gamma\|u_m\|_H^2 + C\|f\|_{V'}^2.$$

Gronwall and integration bound u_m in $L^\infty H \cap L^2 V$. For arbitrary $v\in V$, since $u_m' \in V_m$,

$$\langle u_m', v\rangle = (u_m', P_m v) = \langle f, P_m v\rangle - B_t[u_m, P_m v].$$

Thus $\|u_m'\|_{V'} \le \|f\|_{V'} + M\|u_m\|_V$, uniformly in m. Stability of P_m in V is essential here.

Extract weak limits $u_m \rightharpoonup u$ in L^2V and $u_m' \rightharpoonup z$ in L^2V'. For $\phi \in C_c^\infty(0,T)$, $v\in V$, integration by parts identifies $\int\langle z,v\rangle\phi = -\int (u,v)\phi'$, hence $z = u'$. For each fixed k, the equation tested with $\phi(t)w_k$ passes to the limit. Bounded measurable coefficients act as bounded linear maps on these spaces; weak convergence suffices. Density extends the identity to V. Separability permits one common full-measure set of times.

To recover the initial value take $\phi\in C^1([0,T])$, $\phi(T)=0$, and first $v\in V_N$, $m\ge N$. Then

$$\int_0^T \{-(u_m,v)\phi' + B_t[u_m,v]\phi\}\,dt = \int_0^T \langle f, v\rangle\phi\,dt + (P_m g, v)\phi(0).$$

初值不要求零迹

$g\in L^2(U)$ 无须有边界迹。零边界在正时间的 $L^2(0,T;H_0^1)$ 意义下成立；若要在角点 $\partial U\times\{0\}$ 光滑，才需要相容条件。

Pass to the limit and compare with the integration-by-parts identity for u, whose H trace exists by Lemma 7.1. Thus $(u(0), v) = (g, v)$ on a dense set, and $u(0) = g$. Lower semicontinuity gives the integral estimates. The uniform-in-time bound follows by testing on measurable time sets (Problem 6), followed by H continuity. Finally apply (7.3) to a difference of solutions and repeat the estimate. Zero data force zero difference. □

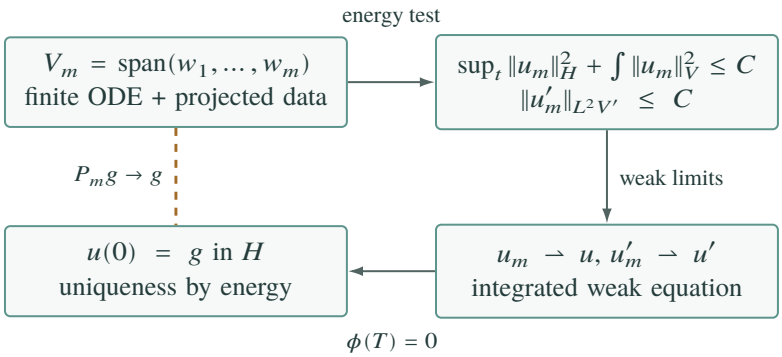


Diagram T7.1. Galerkin approximation preserves the energy estimate. The Dirichlet spectral projection is stable in H_0^1, controlling the time derivative in H^{-1}. Integrated tests retain the initial value in the limit. Arrows describe proof steps, not numerical error bounds.

7.1.3 Improved regularity and compatibility at the corner

Here assume A, b, c are smooth on $\overline{U}$ and independent of time. Write $B_0[u, v] = \int_U a^{ij} u_i v_j$ and $B = B_0 + B_1$; then $|B_1[u, v]| \le C\|u\|_V \|v\|_H$.

Theorem 7.4 (Improved regularity; source §7.1.3, Theorem 5)
If $g \in V$ and $f \in L^2(0, T; H)$, then

$$u \in L^\infty V \cap L^2 H^2(U), \qquad u' \in L^2 H,$$

with norm bound $C(\|g\|_V + \|f\|_{L^2H})$. If also $g \in H^2(U)$ and $f \in H^1(0, T; H)$, then

$$u \in L^\infty H^2, \quad u' \in L^\infty H \cap L^2 V, \quad u'' \in L^2 V',$$

with bound $C(\|g\|_{H^2} + \|f\|_{H^1(0,T;H)})$. All unlabeled time spaces here and below use the interval $(0, T)$.

Expanded proof (legal tests and reconstruction of a time derivative).

Test the finite system with $u_m' \in V_m$. Symmetry and time indepen-

弱极限与初值

L^2 时间弱收敛不控制 $t = 0$。时间分部积分保留 $(g, v)\phi(0)$，再与极限方程比较，才恢复初值；不能直接把 $u_m(0)$ 换成 $u(0)$。

dence give

$$\|u_m'\|_H^2 + \tfrac{1}{2}\tfrac{d}{dt}B_0[u_m,u_m] = (f,u_m') - B_1[u_m,u_m'] \\ \le \tfrac{1}{2}\|u_m'\|_H^2 + C(\|f\|_H^2 + \|u_m\|_V^2).$$

The initial projections are bounded in V; the basic energy estimate bounds $\int \|u_m\|_V^2$. Weak limits yield $u' \in L^2H$ and $u \in L^\infty V$. For a.e. t, $Lu = f - u' \in H$; the elliptic boundary estimate of §6.3 yields

$$\|u(t)\|_{H^2} \le C(\|f(t)\|_H + \|u'(t)\|_H + \|u(t)\|_H).$$

Integrate to prove the first assertion. For the second, construct v from $v' + Lv = f'$, $v(0) = g_1 = f(0) - Lg \in H$. Set $U(t) = g + \int_0^t v(s)\,ds$ in V. The integrated equation in V' gives $v(t) + LU(t) - f(t) = g_1 + Lg - f(0) = 0$. Thus $U = u$ by uniqueness and $u' = v$. The weak estimate bounds v in $L^\infty H \cap L^2 V$, v' in L^2V'. Since $f \in C([0,T];H)$, the elliptic estimate gives $u \in L^\infty H^2$. This argument avoids differentiating a weak solution prematurely. □

Theorem 7.5 (Higher regularity; source Theorems 6–7)
For an integer $m \ge 0$, suppose

$$g \in H^{2m+1}(U), \quad \partial_t^k f \in L^2H^{2m-2k}(U) \quad (0 \le k \le m).$$

Define $g_0 = g$, $g_j = \partial_t^{j-1} f(0) - Lg_{j-1}$ for $1 \le j \le m$. If $g_j \in V$ for $0 \le j \le m$, then

$$\sum_{k=0}^{m+1} \|\partial_t^k u\|_{L^2H^{2m+2-2k}} \le C\left(\|g\|_{H^{2m+1}} + \sum_{k=0}^{m} \|\partial_t^k f\|_{L^2H^{2m-2k}}\right). \tag{7.6}$$

Smooth data satisfying every compatibility condition give $u \in C^\infty(\overline{U} \times [0,T])$.

Expanded proof (the two-derivative induction).

The case $m = 0$ is the preceding theorem. In the step from m to $m+1$, reconstruct $v = u'$ as above. Its compatibility sequence is $g_1, \dots, g_{m+1}$, so induction controls every positive time derivative of u. Recover the missing term from $Lu = f - u'$ and

$$\|u\|_{H^{2m+4}} \le C(\|f - u'\|_{H^{2m+2}} + \|u\|_H).$$

The Sobolev time trace estimate from §5.9, using spatial extension, reads

$$\|F(0)\|_{H^{r+1}} \le C(\|F\|_{L^2H^{r+2}} + \|F'\|_{L^2H^r}) \quad (r \ge 0).$$

系数的时间独立性

若 $A = A(x,t)$，主部能量求导多出 $\frac{1}{2}\int a_t^{ij} u_i u_j$。仅有可测时间依赖时不能使用此计算；基本存在性仍允许时间依赖。

It controls the initial data at each induction step. Combining estimates proves the assertion. For smooth data apply the result for every m; all mixed derivatives then have arbitrarily high Sobolev order on the cylinder. Extension and embedding give continuous derivatives up to its boundary, including the initial corner. □

7.1.4 Weak maximum principles and comparison

Use the nondivergence operator $Pu = u_t - a^{ij}u_{ij} + b^i u_i + cu$, with continuous coefficients on the closed cylinder, symmetric uniformly positive A, and classical $u \in C^{2,1}(U \times (0,T]) \cap C(\overline{U} \times [0,T])$. The parabolic boundary is

$$\Gamma_T = (\overline{U} \times \{0\}) \cup (\partial U \times [0,T]).$$

The top interior $U \times \{T\}$ is not part of Γ_T.

Theorem 7.6 (Weak maximum principle; source Theorems 8–9)
If $Pu \le 0$ and $c \ge 0$, then $\max_{\overline{Q}_T} u \le \max_{\Gamma_T} u^+$. When $c = 0$ the positive part may be removed. Changing signs gives the minimum principle. Hence $Pu \le Pv$, $u \le v$ on Γ_T imply $u \le v$ when $c \ge 0$.

Expanded proof (including a maximum on the final slice).

Set $M = \max_{\Gamma_T} u^+$ and $w = u - M - \varepsilon t$. Then $Pw \le -cM - \varepsilon - c\varepsilon t < 0$. At a positive maximum outside Γ_T, one would have $Dw = 0$, $D^2 w \le 0$, and $w_t \ge 0$ (the left derivative at $t = T$), giving $Pw \ge 0$, a contradiction. Let $\varepsilon \downarrow 0$. When $c = 0$ any constant M may be subtracted. Apply the result to $u - v$ for comparison. If c is only bounded below, write $u = e^{\lambda t} z$; the new potential is $c + \lambda \ge 0$ for large λ. Sign preservation and uniqueness therefore still hold. □

7.1.5 Harnack's inequality and positivity in earlier times

We prove the smooth principal-part case of the source's Harnack theorem: $Pu = u_t - a^{ij}(x,t)u_{ij}$ with smooth symmetric uniformly positive A. A general measurable-coefficient theorem is not proved here.

相容条件是角点条件

$u = 0$ 在侧边界上迫使 $u_t = 0$，而方程给出 $u_t(0) = f(0) - Lg$。热方程 $g = 0, f = 1$ 的一阶相容失败，不能在整个闭柱上声称光滑。

时间方向决定边界

末时刻的内点仍受最大值原理约束。指数代换 $u = e^{\lambda t} z$ 把零阶系数改为 $c + \lambda$，符号不可写反。

Theorem 7.7 (Harnack inequality; source Theorem 10, smooth principal case)
Let smooth $u \geq 0$ solve $Pu = 0$. If $V \Subset U$ is open and connected and $0 < t_1 < t_2 \leq T$, then

$$\sup_V u(\cdot, t_1) \leq C \inf_V u(\cdot, t_2).$$

The constant depends on the interior sets and connecting paths, times, ellipticity, and local coefficient derivative bounds, not on u.

Expanded proof (logarithmic estimate, including cutoff remainders).

First use $u + \varepsilon$ and set $v = \log(u + \varepsilon)$, $p = Dv$, $H_v = D^2 v$, $z = a^{ij} p_i p_j$. Then $v_t = a^{ij} v_{ij} + z$. Put $\mathscr{D} = \partial_t - a^{kl}\partial_{kl} - 2a^{kl} p_l \partial_k$. Differentiating the equation in time gives

$$\mathscr{D} v_t = a_t^{ij}(v_{ij} + p_i p_j).$$

Differentiation of z gives the explicit identity

$$\begin{aligned}\mathscr{D} z = {} & (a_t^{ij} - a^{kl} a_{kl}^{ij}) p_i p_j + 2a^{ij} p_i a_j^{kl} v_{kl} \\ & - 4a^{kl} a_k^{ij} v_{il} p_j - 2a^{ij} a^{kl} v_{ik} v_{jl}.\end{aligned}$$

Here $a_k^{ij} = \partial_k a^{ij}$. The cubic terms in p cancel by symmetry and relabeling the dummy indices. Thus all remaining errors are bounded by $C(|H_v| + |p|^2 + |p||H_v|)$. Choose, for example, $\kappa = 1/2$, and set $F = v_t - (1 - \kappa) z = a^{ij} v_{ij} + \kappa z$. Ellipticity and Young's inequality now give

$$\mathscr{D} F \geq c_0 |H_v|^2 - C|p|^2 - C \qquad (c_0 > 0).$$

Choose a smooth cutoff $0 \leq \zeta \leq 1$, vanishing near $t = 0$ and the spatial boundary, equal to one on a fixed interior ball during $[t_1, t_2]$. At a negative minimum of $\zeta^4 F + \mu t$, $F < 0$ and $DF = -4F D\zeta / \zeta$. Also $F < 0$ implies $|p|^2 \leq C|H_v|$, $|F| \leq C|H_v|$. Applying $\mathscr{D}$ at that minimum produces cutoff errors at most

$$C\zeta^2 |F| + C\zeta^3 |p||F| \leq \delta \zeta^4 |H_v|^2 + C_\delta.$$

The term $C\zeta^4 |p|^2$ is absorbed likewise. A small δ and then large μ contradict $\mathscr{D}(\zeta^4 F + \mu t) \leq 0$. Consequently $F \geq -C$ on the smaller cylinder, and

$$v_t \geq a_0 |Dv|^2 - C_0 \qquad (a_0 > 0). \tag{7.7}$$

Along a path $\eta : [t_1, t_2] \to U$ in that cylinder,

$$\frac{d}{dt} v(\eta(t), t) \geq a_0 |Dv|^2 + Dv \cdot \eta' - C_0 \geq -\frac{|\eta'|^2}{4a_0} - C_0.$$

Integrate along the straight path in a ball and exponentiate. For general V, connect a finite ball covering of its compact closure by finitely many interior paths, divide the time gap into positive increments, and concatenate the estimates. Let $\varepsilon \downarrow 0$; the constants are independent of it. □

Harnack 的时间方向

早时刻的上界受晚时刻的下界控制。时间差趋零时常数通常发散，不能把它读成同一时间切片上的椭圆 Harnack 估计。

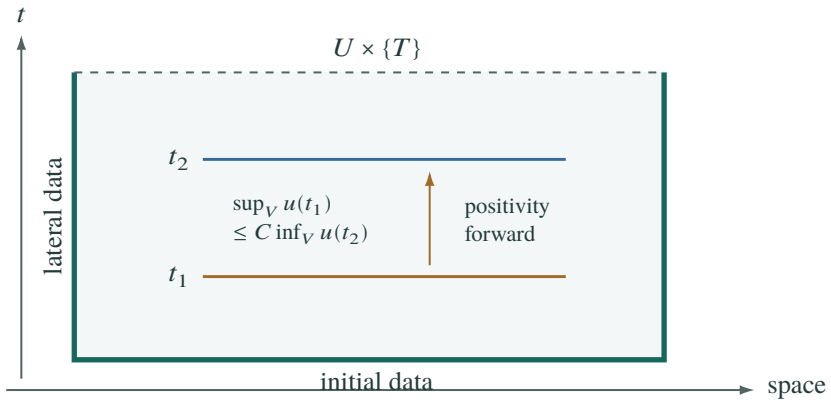


Diagram T7.2. The parabolic boundary and Harnack comparison. The thick bottom and side edges form Γ_T. The top interior belongs to the evolution region. Harnack compares an earlier interior slice to a later one; the arrow indicates forward propagation of positivity.

Theorem 7.8 (Strong maximum principle; source Theorems 11–12, smooth case)

Assume additionally that A is time independent. For $P = \partial_t - a^{ij}\partial_{ij}$ as above on connected U, if $Pu \le 0$ and u attains its global maximum M at $(x_0, t_0) \in U \times (0, T]$, then $u = M$ on $U \times [0, t_0]$. The minimum version follows by changing sign. For $P + c$, $c \ge 0$, the conclusion holds at a nonnegative maximum.

Expanded proof (a local comparison argument).

Set $w = M - u \ge 0$. For the principal case $Pw \ge 0$; in the case $c \ge 0$, $M \ge 0$, one has $(P + c)w \ge cM \ge 0$. Suppose $w(y, s) > 0$ for some $s < t_0$. Choose a smooth connected $W \Subset U$ containing y, x_0, and a nonzero smooth $0 \le \psi \le w(\cdot, s)$ compactly supported near y. Solve $Ph = 0$ on $W \times (s, t_0]$, with zero lateral data and $h(s) = \psi$. This solution is supplied by the preceding existence and regularity theory; compact support ensures corner compatibility. Let $C \ge \sup c$ (take $C = 0$ in the principal case). Then $\tilde{h} = e^{-C(t-s)}h$ satisfies $(P + c)\tilde{h} = (c - C)\tilde{h} \le 0$. Comparison gives $w \ge \tilde{h} \ge 0$. By continuity h is positive somewhere at a time just after s; Harnack then gives $h(x_0, t_0) > 0$, contradicting $w(x_0, t_0) = 0$. Thus $w = 0$ at all earlier interior times; continuity supplies the endpoints. This proof covers the smooth principal coefficients

just treated, with an additional smooth nonnegative potential, not the full rough theory. □

7.2 Second-Order Hyperbolic Equations

7.2.1 Two initial data and the natural energy

Keep $V = H_0^1(U)$, $H = L^2(U)$ and the divergence-form $L(t)$ from §7.1. Here assume $a^{ij}, b^i, c \in C^1(\overline{Q}_T)$, $a^{ij} = a^{ji}$, with the same uniform ellipticity bound. Study

$$u'' + L(t)u = f, \quad u|_{\partial U} = 0, \quad u(0) = g,\ u'(0) = h, \tag{7.8}$$

where $g \in V$, $h \in H$, $f \in L^2(0,T;H)$. A weak solution has

$$u \in L^2V, \quad u' \in L^2H, \quad u'' \in L^2V', \qquad \langle u'', v\rangle + B_t[u,v] = (f,v).$$

The initial values are meaningful in H and V', respectively: $u \in H^1(0,T;H) \subset C([0,T];H)$ and $u' \in H^1(0,T;V') \subset C([0,T];V')$. The estimates below in fact put u, u' in $L^\infty V, L^\infty H$.

Theorem 7.9 (Energy bound and existence; source §7.2.2, Theorems 1–3)
There exists a weak solution satisfying

$$\operatorname*{ess\,sup}_{0<t<T}(\|u(t)\|_V^2 + \|u'(t)\|_H^2) + \|u''\|_{L^2V'}^2 \le C(\|g\|_V^2 + \|h\|_H^2 + \|f\|_{L^2H}^2). \tag{7.9}$$

Expanded proof (the second-order finite system).

Use the same V_m and solve

$$(u_m'', w_k) + B_t[u_m, w_k] = (f, w_k), \quad u_m(0) = P_m g, \quad u_m'(0) = P_m h.$$

This finite second-order system becomes a linear first-order ODE in (d, d'), so has a unique solution. Test with u_m' and put

$$E_m = \tfrac{1}{2}(\|u_m'\|_H^2 + \int_U a^{ij}(u_m)_i(u_m)_j + \|u_m\|_H^2).$$

A direct calculation gives

$$E_m' = (f, u_m') - \int_U (b^i(u_m)_i + cu_m)u_m' + \tfrac{1}{2}\int_U a_t^{ij}(u_m)_i(u_m)_j + (u_m, u_m').$$

Thus $E_m' \le CE_m + C\|f\|_H^2$. Gronwall gives the uniform energy bound; E_m is equivalent to $\|u_m\|_V^2 + \|u_m'\|_H^2$. Using P_m in an arbitrary test gives $\|u_m''\|_{V'} \le C\|u_m\|_V + C\|f\|_H$. Weak compactness yields the equation and

强原理只追溯过去

在 t_0 取内点极值只能推出直到 t_0 的常值性。之后的边界输入或外力可能变化，不能据此断言整个未来为常值。

位移和速度的不同空间

能量包含 $\|Du\|_2^2 + \|u_t\|_2^2$，故自然数据为 $g \in H_0^1$、$h \in L^2$。基本弱解定义中速度初值先在 H^{-1} 识别，不能无证明地宣称 u_t 在 L^2 连续。

identifies successive weak derivatives by scalar time tests, exactly as in the parabolic construction.

For $v \in V_N$, $m \geq N$, and $\phi(T) = \phi'(T) = 0$, twice integrating in time gives

$$\int_0^T \{(u_m, v)\phi'' + B_t[u_m, v]\phi\}\, dt = \int_0^T (f, v)\phi\, dt + (P_m h, v)\phi(0) - (P_m g, v)\phi'(0).$$

Both endpoint coefficients survive the weak limit. Comparing with the trace formula for u, u' identifies $u(0) = g$, $u'(0) = h$. Lower semicontinuity, including the time-set argument of Problem 6, gives (7.9). □

7.2.2 Uniqueness without testing by an unavailable derivative

A weak solution need not have $u' \in V$, so the formal test $v = u'$ is not legitimate. The source's integrated test repairs this gap.

Theorem 7.10 (Weak uniqueness; source Theorem 4)
The weak solution of (7.8) *is unique in the class just defined.*

Expanded proof (the integrated test).

Subtract two solutions, so $f = g = h = 0$. Fix s and use $v(t) = \int_t^s u(r)\, dr$ for $0 \leq t \leq s$. Then $v' = -u$ in V, $v(s) = 0$, and v is a legal time-dependent V test by approximation with finite sums of scalar time tests. Write $B_t = S_t + K_t$, with S_t symmetric and K_t skew-symmetric. Integration by parts in space gives

$$S_t[z, z] = \int_U a^{ij} z_i z_j + (c - \tfrac{1}{2}\operatorname{div} b) z^2,$$
$$K_t[u, v] = -\int_U u\, b \cdot Dv - \tfrac{1}{2}(\operatorname{div} b) uv.$$

In particular $|K_t[u, v]| \leq C\|u\|_H \|v\|_V$, $|\dot{S}_t[v, v]| \leq C\|v\|_V^2$ (one may use the original symmetric formula, avoiding mixed coefficient derivatives). Also $S_t[v, v] \geq \alpha\|v\|_V^2 - \gamma\|v\|_H^2$. Time integration by parts, with $u'(0) = 0$ in V', gives

$$\tfrac{1}{2}\|u(s)\|_H^2 + \tfrac{1}{2} S_0[v(0), v(0)] = -\int_0^s (\tfrac{1}{2}\dot{S}_t[v, v] + K_t[u, v])\, dt.$$

Here $\int (u', u) = \frac{1}{2}\|u(s)\|_H^2$ is valid since $u \in H^1(0, T; H)$; all other pairings involve V', V. Thus

$$\|u(s)\|_H^2 + \alpha\|v(0)\|_V^2 \leq C \int_0^s (\|u\|_H^2 + \|v\|_V^2)\, dt + \gamma\|v(0)\|_H^2.$$

Put $W(t) = \int_0^t u$, so $v(t) = W(s) - W(t)$. The last term is bounded by $\gamma s \int_0^s \|u\|_H^2$. Consequently, for a fixed sufficiently short interval,

$$\|u(s)\|_H^2 + \tfrac{\alpha}{2}\|W(s)\|_V^2 \leq C \int_0^s (\|u(t)\|_H^2 + \|W(t)\|_V^2)\, dt.$$

Gronwall forces $u = 0$ there. Restart at its right endpoint: the H and V' traces are zero, and all constants are uniform on $[0, T]$. Finitely many repetitions give $u = 0$ throughout. □

7.2.3 Higher regularity and the absence of smoothing

For the remainder of the bounded-domain regularity discussion, coefficients are smooth and time independent.

Theorem 7.11 (Improved wave regularity; source §7.2.3, Theorem 5)
If $g \in H^2(U) \cap V$, $h \in V$, and $f \in H^1(0, T; H)$, then

$$u \in L^\infty H^2, \quad u' \in L^\infty V, \quad u'' \in L^\infty H, \quad u''' \in L^2 V',$$

with norm bound $C(\|g\|_{H^2} + \|h\|_V + \|f\|_{H^1(0,T;H)})$.

Expanded proof (reconstruction instead of formal differentiation).

Construct v solving $v'' + Lv = f'$ with initial data $v(0) = h$, $v'(0) = f(0) - Lg \in H$. Set $U = g + \int_0^t v$. In distributions, $(U'' + LU - f)' = v'' + Lv - f' = 0$, while the residual at 0 is zero in V'. Hence U solves the original equation and has data g, h. Uniqueness gives $U = u$, $u' = v$. Estimate v by (7.9) and recover $u \in L^\infty H^2$ from $Lu = f - u''$ and the elliptic estimate. This proves every stated norm bound. □

Theorem 7.12 (Higher wave regularity; source Theorems 6–7)
Let $m \geq 0$, $g \in H^{m+1}(U)$, $h \in H^m(U)$, and $\partial_t^k f \in L^2 H^{m-k}(U)$ for $0 \leq k \leq m$. Set $q_0 = g$, $q_1 = h$, and

$$q_{j+2} = \partial_t^j f(0) - Lq_j \quad (0 \leq j \leq m-2).$$

Require $q_j \in V$ for $0 \leq j \leq m$ (for $m = 0$ only $g \in V$). Then

$$\operatorname*{ess\,sup}_{0<t<T} \sum_{k=0}^{m+1} \|\partial_t^k u(t)\|_{H^{m+1-k}} \leq C\left(\|g\|_{H^{m+1}} + \|h\|_{H^m} + \sum_{k=0}^{m} \|\partial_t^k f\|_{L^2 H^{m-k}}\right).$$

Smooth data compatible to every order give a smooth solution on the closed cylinder.

Expanded proof (one time derivative at each induction step).

The case $m = 0$ is the basic wave bound. In passing to $m + 1$, the preceding reconstruction gives $v = u'$ with data $h, f(0) - Lg$ and forcing

积分测试恢复合法性

$v(t) = \int_t^s u$ 属于 H_0^1，而 $v' = -u$。它将不可用的 u_t 测试转成合法的时间分部积分；短时间吸收后再拼接，覆盖整个时间区间。

f'. Its compatibility sequence is shifted by one place. Apply induction to v. The trace inequality $\|F(0)\|_{H^r} \le C(\|F\|_{L^2H^r} + \|F'\|_{L^2H^r})$ controls the forcing traces required at this step. Then $Lu = f - u''$ and elliptic regularity recover the missing highest spatial norm of u. This proves the estimate and the induction. Arbitrarily high mixed Sobolev regularity and embedding prove the smooth conclusion. □

7.2.4 Finite propagation and the geometry of the cone

On $\mathbb{R}^n$ now take the source's nondivergence model $u_{tt} - a^{ij}(x)u_{ij} = 0$ with smooth symmetric positive A. Assume on the region under consideration that an optical distance q is smooth away from x_0, positive there, $q(x_0) = 0$, and

$$ADq \cdot Dq = 1.$$

Assume its relevant sublevel sets are bounded and shrink to x_0. Set $K_t = \{q < t_0 - t\}$ for $0 \le t < t_0$ and $K = \{(x,t) : 0 \le t < t_0,\ x \in K_t\}$.

> **Theorem 7.13** (Finite propagation; source §7.2.4, Theorem 8)
> *If a smooth solution has $u = u_t = 0$ on K_0, it vanishes throughout K, and at its vertex by continuity.*

Expanded proof (coarea, normal speed, and the energy flux).

The outward normal is $\nu = Dq/|Dq|$ and the moving boundary has outward normal velocity $-1/|Dq|$. Hence for $E(t) = \frac{1}{2}\int_{K_t}(u_t^2 + ADu\cdot Du)$, differentiation gives

$$E'(t) = -\int_{K_t} u_t(\partial_j a^{ij})u_i + \int_{\partial K_t} u_t ADu\cdot\nu - \frac{1}{2}\int_{\partial K_t}\frac{u_t^2 + ADu\cdot Du}{|Dq|}.$$

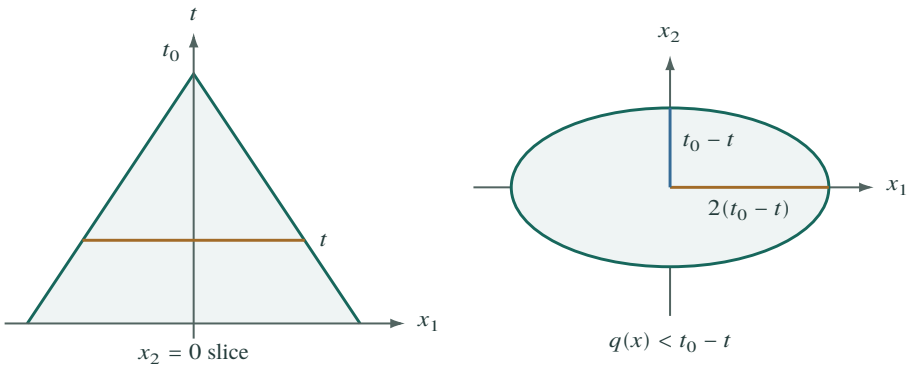
The volume error is at most $CE(t)$ by ellipticity and bounded first coefficient derivatives. Also

$$|ADu\cdot\nu| \le (ADu\cdot Du)^{1/2}(A\nu\cdot\nu)^{1/2} = \frac{(ADu\cdot Du)^{1/2}}{|Dq|}.$$

Young's inequality makes the boundary flux no larger than the negative moving-boundary term. Thus $E' \le CE$ and $E(0) = 0$, giving $E = 0$. For fixed $x \in K_t$, that same x belongs to every earlier K_s; integrating $u_t = 0$ from its zero initial value proves $u(x,t) = 0$. □

波动不会自动磨光

$\sin(kx)$ 模态的位移为 $\cos(kt)\sin(kx)$；高频振幅没有热方程的 e^{-k^2t} 衰减。提高解的正则性需要提高初值正则性，不能从能量估计推导任意正时刻光滑。



光学距离的局部限制

度量是 A^{-1}，不是 A。变系数的距离函数可能有割迹，不能默认全空间光滑；此处光滑 q 是明确假设。常系数时 $q = \sqrt{(x-x_0)^T A^{-1}(x-x_0)}$，波前为椭球。

Diagram T7.3. Anisotropy changes the cross-section of a wave cone. For $A = \operatorname{diag}(4,1)$, $q(x) = \sqrt{x_1^2/4 + x_2^2}$. At time t, the dependency cross-section is an ellipse with semiaxes $2(t_0 - t)$ and $t_0 - t$. The cone sketch and the spatial slice use separate coordinate scales.

7.2.5 Characteristic coordinates in two variables

For $au_{xx} + 2bu_{xy} + du_{yy} + \text{lower terms} = 0$, assume smooth coefficients and $ad - b^2 < 0$. A coordinate function ξ must solve

$$a\xi_x^2 + 2b\xi_x\xi_y + d\xi_y^2 = 0.$$

After a fixed linear coordinate change one can arrange $a \neq 0$ near the point: an indefinite quadratic form cannot vanish on every direction. Writing $D = b^2 - ad > 0$ factors this equation into the two distinct fields

$$a\xi_x + (b + \sqrt{D})\xi_y = 0, \qquad a\eta_x + (b - \sqrt{D})\eta_y = 0.$$

Expanded proof (local invertibility and both canonical forms).

The two smooth nonzero vector fields are transverse, since their determinant is $-2a\sqrt{D} \neq 0$. The local flow-box construction (solving the characteristic ODE from a transverse curve) gives first integrals ξ, η with nonzero independent gradients. Thus $\Phi = (\xi, \eta)$ is locally invertible. By the chain rule the transformed principal matrix is $\widetilde{A} = D\Phi\, A\, D\Phi^T$, whose two diagonal entries vanish. Its determinant is $(\det D\Phi)^2 \det A < 0$, so the off-diagonal entry is nonzero. Division yields $v_{\xi\eta} + \text{lower terms} = 0$. For $z_1 = \xi + \eta$, $z_2 = \xi - \eta$, this becomes $v_{z_1 z_1} - v_{z_2 z_2} + \text{lower terms} = 0$. For example $u_{tt} - c^2 u_{xx}$, $c > 0$, becomes $-4c^2 v_{\xi\eta}$ with $\xi = x - ct$, $\eta = x + ct$. All these coordinates are local; no global characteristic chart is asserted. □

7.3 Systems of Hyperbolic First-Order Equations

7.3.1 The symbol and the meaning of hyperbolicity

双曲标准形是局部结论

判别式给出两个横截特征方向；逆函数定理还需要 $\det D\Phi \neq 0$。先处理 $a \neq 0$ 的局部坐标，避免在 $a = 0$ 处直接除法。

For vector-valued $u : \mathbb{R}^n \times [0,T] \to \mathbb{R}^m$ study

$$u_t + \sum_{j=1}^{n} B^j(x,t)u_{x_j} = f, \qquad u(0) = g. \tag{7.10}$$

Its spatial symbol is $B(x,t;\xi) = \sum_j \xi_j B^j(x,t)$. Following the source, the system is hyperbolic if this matrix has a basis of real eigenvectors and real eigenvalues for each nonzero ξ; strictly hyperbolic means all eigenvalues are real and distinct. It is symmetric hyperbolic if every B^j is real symmetric. A plane wave $u = r\,F(\xi \cdot x - \sigma t)$ solves the constant coefficient homogeneous equation if $B(\xi)r = \sigma r$. For $|\xi| = 1$, σ is its signed speed. Repeated eigenvalues are permitted in symmetric systems.

More generally, $B^0 u_t + \sum B^j u_{x_j} = f$ is symmetrizable in this energy argument when all B^j, including B^0, are symmetric and B^0 is uniformly positive. Its energy is $\int u \cdot B^0 u$. For $v_{tt} - a^{ij}v_{ij} = 0$ set $u = (Dv, v_t)$, $B^0 = \operatorname{diag}(A, 1)$, and

$$B^j = \begin{pmatrix} 0 & -Ae_j \\ -e_j^T A & 0 \end{pmatrix}.$$

实谱与一致对角化

每个方向可对角化，并不自动给出方向上一致有界的对角化矩阵。L^2 适定性需要额外的一致控制；对称矩阵可用正交对角化，条件数恒为 1。

This gives a symmetric first-order system, since $(Dv)_t = Dv_t$ and $v_{tt} = a^{ij}v_{ij}$. Recovering v also requires the initial gradient constraint; arbitrary first-order data need not be gradients.

7.3.2 Symmetry, energy, and vanishing viscosity

Assume $B^j \in C_b^2(\mathbb{R}^n \times [0,T]; \mathbb{R}^{m\times m})$ are symmetric, $g \in H^1(\mathbb{R}^n;\mathbb{R}^m)$, and $f \in H^1(\mathbb{R}^n \times (0,T);\mathbb{R}^m)$. Here C_b^2 means that the matrices and all derivatives through order two are bounded. For smooth functions, spatial integration by parts yields

$$\frac{1}{2}\frac{d}{dt}\|u\|_2^2 = (f,u) + \frac{1}{2}\int_{\mathbb{R}^n} u \cdot \Big(\sum_j \partial_j B^j\Big)u. \tag{7.11}$$

The identity follows from $2u\cdot B^j u_j = \partial_j(u\cdot B^j u) - u\cdot(\partial_j B^j)u$. For Sobolev functions on $\mathbb{R}^n$, insert expanding compact cutoffs; the errors vanish by L^2 integrability and bounded coefficients.

Theorem 7.14 (Symmetric hyperbolic well-posedness; source Theorems 1–4) *There is a unique solution of (7.10) with $u \in L^2(0,T;H^1)$, $u_t \in L^2(0,T;L^2)$, $u(0) = g$ in L^2. It satisfies*

$$\|u\|_{L^\infty H^1} + \|u_t\|_{L^2L^2} \le C(\|g\|_{H^1} + \|f\|_{H^1(\mathbb{R}^n\times(0,T))}).$$

The weak identity uses H^1 tests and the ordinary L^2 inner product.

Expanded proof (viscosity construction).

For $0 < \varepsilon \le 1$ solve

$$u_t^\varepsilon - \varepsilon\Delta u^\varepsilon + \sum_j B^j u_j^\varepsilon = f, \qquad u^\varepsilon(0) = g_\varepsilon = \rho_\varepsilon * g.$$

For fixed ε, Duhamel's formula with the heat semigroup gives an elementary construction. The estimate $\|De^{\varepsilon r\Delta}h\|_2 \le C(\varepsilon r)^{-1/2}\|h\|_2$ follows from its Fourier multiplier. Thus the integral operator is a contraction on $C([0,\delta];H^1)$ when $C(\delta + \sqrt{\delta/\varepsilon}) < 1$. Iterate finitely many intervals; the forcing term is continuous in H^1 because $f \in L^1(0,T;H^1)$. Smooth approximations to coefficients and data justify the following computations; their bounds depend only on the indicated norms.

Testing with u^ε gives (7.11) with the additional positive term $\varepsilon\|Du^\varepsilon\|_2^2$ on the left. Differentiate in x_k. The new forcing is $f_k - \sum_j (\partial_k B^j)u_j^\varepsilon$, bounded by $\|Df\|_2 + C\|Du^\varepsilon\|_2$. Summing the energy estimates and applying Gronwall yields

$$\sup_t \|u^\varepsilon(t)\|_{H^1}^2 + \varepsilon\int_0^T \|D^2u^\varepsilon\|_2^2\,dt \le C(\|g\|_{H^1}^2 + \|f\|_{L^2H^1}^2).$$

The equation then gives

$$\|u_t^\varepsilon\|_{L^2L^2} \le \varepsilon\|\Delta u^\varepsilon\|_{L^2L^2} + C\|u^\varepsilon\|_{L^2H^1} + \|f\|_{L^2L^2} \le C.$$

In fact the viscosity term tends to zero in L^2L^2 at rate at most $C\sqrt{\varepsilon}$ by the previous bound. Weak compactness now yields $u^\varepsilon \rightharpoonup u$ in L^2H^1 and $u_t^\varepsilon \rightharpoonup u_t$ in L^2L^2. All terms pass to the limit in integrated tests. Taking tests vanishing at T and integrating the time derivative recovers $u(0) = g$; $H^1(0,T;L^2)$ supplies a continuous L^2 representative. Finally use (7.11) for a difference of two solutions. The identity is legal at this regularity by spatial approximation and the Hilbert-valued chain rule. Gronwall proves uniqueness and continuous dependence. □

对称性的精确用途

对称性把 $u \cdot B^j u_j$ 写成通量散度加零阶项。若 B^j 不对称，会留下含 Du 的项，无法仅靠 $\|u\|_2$ 封闭估计。

7.3.3 Constant coefficients: real spectrum and derivative loss

Assume now constant real matrices and only that $B(\xi)$ has real spectrum; it may have Jordan blocks. Fourier transformation in space gives

$$\hat{u}(\xi, t) = e^{-itB(\xi)}\hat{g}(\xi). \tag{7.12}$$

This is a weaker hypothesis than the hyperbolicity definition above.

Lemma 7.15 (Polynomial multiplier bound; source proof of Theorem 5)
For each finite T, $\|e^{-itB(\xi)}\| \le C_T(1 + |\xi|^{m-1})$ for $0 \le t \le T$.

Expanded proof (triangular multiplier bound).

By complex Schur triangularization choose a unitary matrix Q such that $Q^*B(\xi)Q = D + N$, where D is real diagonal and N strictly upper triangular. Both have norm at most $C|\xi|$, uniformly in direction. Variation of constants around D expresses the matrix exponential as an ordered integral series. A product with r factors of N, interspersed with diagonal exponentials, has zero entries below the rth upper diagonal; it therefore vanishes for $r \ge m$. Diagonal exponentials have norm one, and the time simplex has volume $t^r/r!$. Consequently

$$\|e^{-itB(\xi)}\| \le \sum_{r=0}^{m-1} \frac{(Ct|\xi|)^r}{r!}.$$

Unitary conjugation preserves the norm. This proves a direction-uniform bound without assuming uniformly conditioned eigenvectors. □

Theorem 7.16 (Fourier solution; source Theorem 5, specified uniqueness class)
If $g \in H^s(\mathbb{R}^n;\mathbb{R}^m)$, $s > n/2 + m$, the inverse transform of (7.12) is C^1 in (x,t) on every finite time strip and solves the homogeneous system with initial data g. It belongs to $C([0,T];H^{s-m+1})$, with time derivative in $C([0,T];H^{s-m})$, and is unique among solutions with these properties.

Expanded proof (integrability, differentiation, and uniqueness).

The multiplier bound gives the two asserted Sobolev bounds. For pointwise differentiation, both space and time derivatives add at most one factor $C|\xi|$. Cauchy–Schwarz shows

$$\int_{\mathbb{R}^n}(1 + |\xi|)^m|\hat{g}(\xi)|\,d\xi \le C\|g\|_{H^s}\left(\int_{\mathbb{R}^n}(1 + |\xi|)^{-2(s-m)}d\xi\right)^{1/2} < \infty.$$

黏性极限的统一常数

固定 ε 的抛物正则性常数可能随 $\varepsilon \downarrow 0$ 发散。这里另证一致 H^1 估计，并由 $\varepsilon \int \|D^2u^\varepsilon\|_2^2 \le C$ 得到黏性项消失。

Thus dominated convergence proves C^1 regularity and justifies substitution into the equation. Continuity in the stated Sobolev spaces follows from the same multiplier bound. For uniqueness, Fourier localize a difference to a compact frequency set; multiplication by $B(\xi)$ is then a bounded operator on L^2 of that set. The resulting Banach-space ODE with zero initial value is zero. An increasing sequence of compact frequency cutoffs proves uniqueness. No unrestricted uniqueness assertion for arbitrary classical solutions with unspecified growth is needed. □

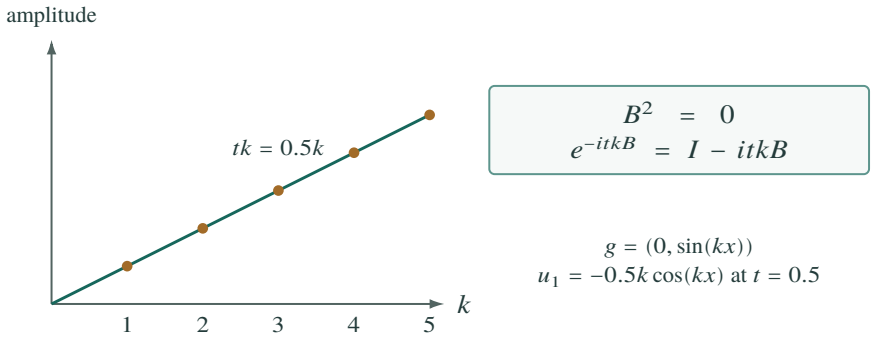


Diagram T7.4. A Jordan block amplifies high frequencies. For $B = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ and input $g = (0, \sin(kx))$ on a periodic line, $u = (-tk\cos(kx), \sin(kx))$. At $t = 0.5$ the first component has amplitude $0.5k$. This exact periodic model illustrates the Fourier multiplier mechanism; the whole-space theorem above uses Sobolev data on $\mathbb{R}^n$.

Jordan 块造成导数损失

对 $B = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$，$u_2 = g_2$、$u_1 = g_1 - tg_2'$。谱全为实数，仍需要多一个空间导数；实谱本身不保证 $L^2 \to L^2$ 有界。

7.4 Semigroup Theory

7.4.1 Strong continuity, the generator, and its domain

Let X be a real Banach space. A strongly continuous semigroup is a family $S(t) \in \mathscr{L}(X)$, $t \geq 0$, satisfying

$$S(0) = I, \quad S(t+s) = S(t)S(s), \quad S(t)x \longrightarrow x \text{ in } X \text{ as } t \downarrow 0.$$

We first assume $\|S(t)\| \leq 1$ (a contraction semigroup). Its generator is the generally unbounded operator

$$D(G) = \left\{ x : \lim_{h \downarrow 0} \frac{S(h)x - x}{h} \text{ exists in } X \right\}, \qquad Gx = \lim_{h \downarrow 0} \frac{S(h)x - x}{h}.$$

We use G instead of the source's A, to distinguish it from the principal coefficient matrix. For $u_t + Lu = 0$, the generator is $G = -L$.

Theorem 7.17 (Generator properties; source §7.4.1, Theorems 1–2)
The domain $D(G)$ is dense, and G is closed. For $x \in D(G)$, $S(t)x \in D(G)$, $GS(t)x = S(t)Gx$, and

$$\frac{d}{dt}S(t)x = GS(t)x = S(t)Gx.$$

At $t = 0$ this is a right derivative. In particular $S(t)x - x = \int_0^t S(s)Gx\,ds$.

Expanded proof (domain invariance, density, and graph closure).

Commuting $S(t)$ with $S(h)$ and passing to the defining difference quotient proves invariance and commutation. The right derivative follows likewise; for the left derivative at $t > 0$ use

$$\frac{S(t)x - S(t-h)x}{h} = S(t-h)\frac{S(h)x - x}{h} \longrightarrow S(t)Gx.$$

Strong continuity and the contraction bound justify the last limit. For arbitrary $x \in X$, let $x_t = t^{-1}\int_0^t S(s)x\,ds$. A change of integration interval gives

$$x_t \in D(G), \qquad Gx_t = \frac{S(t)x - x}{t}, \qquad x_t \to x.$$

Thus the domain is dense. To prove closedness, take $x_k \to x$, $Gx_k \to y$ in X and pass to the limit in $S(t)x_k - x_k = \int_0^t S(s)Gx_k\,ds$. The contraction bound makes convergence of the integrals uniform on bounded time intervals. Divide the limit identity by t and let $t \downarrow 0$: its right side tends to y. Hence $x \in D(G)$ and $Gx = y$. □

7.4.2 Resolvents as Laplace transforms

For a closed densely defined operator G, let $\rho(G)$ be the set of real λ for which $\lambda I - G : D(G) \to X$ is bijective, and $R_\lambda = (\lambda I - G)^{-1}$. The closed graph theorem makes $R_\lambda : X \to X$ bounded. Its range is $D(G)$.

Theorem 7.18 (Resolvent identities and representation; source Theorem 3)
For $\lambda, \mu \in \rho(G)$,

$$R_\lambda - R_\mu = (\mu - \lambda)R_\lambda R_\mu, \qquad R_\lambda R_\mu = R_\mu R_\lambda. \tag{7.13}$$

For a contraction generator every $\lambda > 0$ is in $\rho(G)$, and

$$R_\lambda x = \int_0^\infty e^{-\lambda t}S(t)x\,dt, \qquad \|R_\lambda\| \le \lambda^{-1}. \tag{7.14}$$

The integral is a norm-convergent Bochner integral.

强连续不等于范数连续

要求每个固定 x 的 $\|S(t)x - x\| \to 0$。若要求 $\|S(t) - I\| \to 0$，就排除了许多无界生成元，例如 L^2 上的热半群。

闭算子与有界算子的区别

闭性是 $x_k \to x$ 且 $Gx_k \to y$ 时才推出 $Gx = y$，不表示 $\|Gx\| \le C\|x\|$。热方程的高频特征函数说明后一不等式不成立。

Expanded proof (the inverse on both sides).

For the first identity insert $(\mu I - G) - (\lambda I - G) = (\mu - \lambda)I$ between R_λ and R_μ. These compositions are defined because each resolvent maps into $D(G)$. Interchanging λ, μ gives commutation (equal parameters are immediate). Let $J_\lambda x$ denote the integral in (7.14). It converges since its integrand has norm at most $e^{-\lambda t}\|x\|$. Changing variables gives

$$\frac{S(h)J_\lambda x - J_\lambda x}{h} = \frac{e^{\lambda h} - 1}{h} J_\lambda x - \frac{e^{\lambda h}}{h} \int_0^h e^{-\lambda t} S(t)x \, dt.$$

The limit is $\lambda J_\lambda x - x$. Thus $J_\lambda x \in D(G)$ and $(\lambda I - G)J_\lambda x = x$. For $x \in D(G)$, differentiate $e^{-\lambda t}S(t)x$ and integrate:

$$\int_0^\infty e^{-\lambda t} S(t)\,(\lambda x - Gx)\, dt = x.$$

The endpoint at infinity vanishes by the contraction bound. Therefore J_λ is also a left inverse, proving bijectivity and the formula. The norm bound follows by integration. This proof never passes an unbounded operator through an integral without a domain argument; Problem 13 gives the equivalent graph-norm justification. □

7.4.3 Constructing the semigroup from resolvent bounds

Theorem 7.19 (Hille–Yosida, contraction case; source §7.4.2, Theorem 4)
A closed densely defined linear G generates a contraction semigroup if and only if

$$(0, \infty) \subset \rho(G), \qquad \|R_\lambda\| \le \lambda^{-1} \quad (\lambda > 0).$$

Expanded proof (Yosida approximation).

Necessity is Theorem 7.18. Conversely define the bounded operators

$$G_\lambda = \lambda^2 R_\lambda - \lambda I = \lambda G R_\lambda.$$

For $x \in D(G)$, $\lambda R_\lambda x - x = R_\lambda G x \to 0$. The bound $\|\lambda R_\lambda\| \le 1$ and density extend this convergence to every $x \in X$. Consequently $G_\lambda x = \lambda R_\lambda G x \to Gx$ for $x \in D(G)$. Put $S_\lambda(t) = e^{tG_\lambda}$. Its power series gives

$$\|S_\lambda(t)\| \le e^{-\lambda t} \sum_{k=0}^{\infty} \frac{(\lambda^2 t)^k \|R_\lambda\|^k}{k!} \le 1.$$

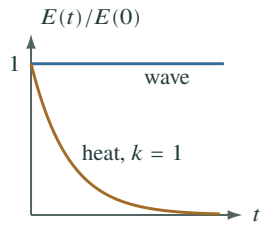
The resolvent identity shows that all G_λ commute. Differentiate $S_\mu(t - s)S_\lambda(s)x$ in s and integrate to obtain

$$\|S_\lambda(t)x - S_\mu(t)x\| \le t\|(G_\lambda - G_\mu)x\| \longrightarrow 0 \quad (x \in D(G)).$$

预解算子的符号

$G = -L$ 时 $R_\lambda = (\lambda I + L)^{-1}$。椭圆能量估计给出的正参数正好对应发展方程的稳定方向；若误写成 $L - \lambda I$，会引入不应出现的共振。

图 T7.M：同一模态的两种衰减



热方程取 $E = \frac{1}{2}\|u\|_2^2$，图示 $k = 1$ 模态。波动方程取 $E = \frac{1}{2}(\|u_t\|_2^2 + \|u_x\|_2^2)$，同一模态总能量守恒。

Thus $S_\lambda(t)x$ converges uniformly on bounded time intervals. Approximation of any $x \in X$ by points of $D(G)$ extends this convergence to all X. The limit $S(t)$ is a contraction, is strongly continuous, and satisfies the semigroup law: in the product limit use the uniform operator bound to control one factor and strong convergence for the other.

Let K be its generator. For $x \in D(G)$ pass to the limit in

$$S_\lambda(t)x - x = \int_0^t S_\lambda(s)G_\lambda x\,ds$$

to get $S(t)x - x = \int_0^t S(s)Gx\,ds$. Dividing by t proves $G \subset K$. This inclusion alone is insufficient. Fix $\lambda > 0$ and $y \in D(K)$, and define $x = R_\lambda(\lambda I - K)y \in D(G)$. Since $Kx = Gx$, $(\lambda I - K)x = (\lambda I - K)y$. The operator $\lambda I - K$ is injective by Theorem 7.18, so $y = x \in D(G)$. Thus $D(K) = D(G)$ and $K = G$. □

For $\|S(t)\| \le e^{\omega t}$, apply the theorem to $G - \omega I$. The resolvent condition becomes $\|R_\lambda\| \le (\lambda - \omega)^{-1}$ for $\lambda > \omega$. The theorem here has leading constant one; a general bound $Me^{\omega t}$ requires the corresponding bounds on all resolvent powers.

不能只证生成元的包含关系

极限积分公式只给出 $G \subset K$。再用共同正参数下的预解算子满射和单射，才能排除 K 有更大定义域，完成 $G = K$。

7.4.4 Parabolic and wave generators

Assume again smooth time-independent elliptic coefficients and smooth bounded U. Set $D(G) = H^2(U) \cap V$, $Gu = -Lu$ on H.

Theorem 7.20 (Parabolic semigroup; source §7.4.3, Theorem 5)
This G generates a semigroup on H with $\|S(t)\| \le e^{\gamma t}$, where $B[u,u] \ge \alpha\|u\|_V^2 - \gamma\|u\|_H^2$.

Expanded proof (the elliptic resolvent and graph estimate).

Density follows from $C_c^\infty(U) \subset D(G)$. If $u_k \to u$ and $Gu_k \to z$ in H, the elliptic estimate

$$\|u_k - u_\ell\|_{H^2} \le C(\|G(u_k - u_\ell)\|_H + \|u_k - u_\ell\|_H)$$

shows convergence in H^2. The trace is still zero and $Gu = z$; thus G is closed. For $\lambda > \gamma$, Lax–Milgram solves $B[u,v] + \lambda(u,v) = (f,v)$, and elliptic regularity puts u in $D(G)$. Testing with u gives $(\lambda - \gamma)\|u\|_H^2 \le \|f\|_H\|u\|_H$. Therefore $\|R_\lambda\| \le (\lambda - \gamma)^{-1}$, and the shifted Hille–Yosida theorem applies. For $g \in D(G)$ the orbit is a classical H-valued solution. For $g \in H$ approximate by $D(G)$ data and use the energy estimate; the semigroup orbit agrees with the weak parabolic solution. Domain regularity is not claimed at $t = 0$ for arbitrary H data. □

For the wave application require $b = 0, A = A^T, c \geq 0$. The symmetric form B is coercive on V. On $X = V \times H$ use

$$\|(u,v)\|_X^2 = B[u,u] + \|v\|_H^2, \qquad \mathcal{G}(u,v) = (v, -Lu),$$

with $D(\mathcal{G}) = (H^2(U) \cap V) \times V$.

Theorem 7.21 (Wave semigroup; source Theorem 6)
The operator $\mathcal{G}$ generates a contraction semigroup on X. In fact it extends to an isometric group.

Expanded proof (wave resolvent).

The domain is dense. Closedness follows as above: convergence of (u_k, v_k) and $(v_k, -Lu_k)$ in $V \times H$ implies $v \in V$ and $u \in H^2 \cap V$, with the correct graph limit. For $\lambda > 0$, solving $(\lambda I - \mathcal{G})(u,v) = (F, H_0)$ is equivalent to

$$(L + \lambda^2)u = H_0 + \lambda F, \qquad v = \lambda u - F.$$

Lax–Milgram and elliptic regularity give the unique domain solution. Moreover $\langle \mathcal{G}(u,v), (u,v)\rangle_X = B[v,u] - (Lu, v) = 0$ by symmetry. Hence

$$\lambda \|(u,v)\|_X^2 = \langle (F, H_0), (u,v)\rangle_X,$$

which proves $\|R_\lambda\| \leq \lambda^{-1}$. Hille–Yosida gives the contraction semigroup. The same argument works for $-\mathcal{G}$. The two semigroups are inverses: on domain data differentiate their product, then use density. Both are contractions, so each preserves the norm. This proves the group assertion as an editorial supplement, and explains energy conservation. □

7.4.5 Forcing and mild solutions: an editorial bridge

For $f \in L^1(0,T;X)$ the variation-of-constants formula is

$$u(t) = S(t)g + \int_0^t S(t-s)f(s)\,ds.$$

It defines a continuous X-valued mild solution; continuity follows from strong continuity and the integrable norm bound. For domain-valued data with continuous graph-norm forcing, differentiation and Theorem 7.17 verify $u' = Gu + f$. Approximation extends the formula as a mild solution for general g, f. Uniqueness for classical domain-valued solutions follows by differentiating $S(t-s)u(s)$. For parabolic forcing in $L^2(0,T;H)$ the energy estimate and approximation identify this formula with the weak solution. A general mild orbit is not asserted to be differentiable for arbitrary Banach-space forcing.

自主与非自主发展

单参数半群依赖系数不含时间。$L = L(t)$ 时通常要用双参数传播族 $U(t,s)$；不能把 $e^{-tL(t)}$ 当成解算子。Galerkin 法仍可直接处理此情形。

波动的压缩实际是等距

无阻尼、对称且时间独立时，反向演化也压缩，因此正向演化只能保持能量。不能由“压缩半群”一词推断波的能量严格衰减。

7.5 Problems and Worked Solutions

The following are concise restatements of all 16 source exercises, with editorial solutions. Unless specified otherwise, U is bounded with smooth boundary and all classical solutions have enough regularity for the indicated integrations. The source problem numbers are retained in the paragraph labels.

1. Neumann heat uniqueness. Prove uniqueness for $u_t - \Delta u = f$, $\partial_\nu u = 0$, $u(0) = g$.

Solution. A difference w has zero initial value and homogeneous equation and boundary data. Multiplication by w yields $\frac{1}{2}\frac{d}{dt}\|w\|_2^2 + \|Dw\|_2^2 = 0$; the boundary term is $\int_{\partial U} w\partial_\nu w = 0$. Thus $w = 0$. Neumann constants do not obstruct evolutionary uniqueness once the initial value is fixed. □

2. Sharp Dirichlet heat decay. Show $\|u(t)\|_2 \le e^{-\lambda_1 t}\|g\|_2$ for homogeneous heat flow.

Solution. The energy identity and the Rayleigh characterization from §6.5 give $\frac{d}{dt}\|u\|_2^2 = -2\|Du\|_2^2 \le -2\lambda_1\|u\|_2^2$. Multiply by $e^{2\lambda_1 t}$, integrate, then take square roots. Equality holds for a first Dirichlet eigenfunction; this fixes the factor two. □

平方范数的衰减率

$\|u\|_2^2$ 以 $2\lambda_1$ 衰减，而 $\|u\|_2$ 以 λ_1 衰减。Neumann 问题有常数模态，不能直接使用正的 Dirichlet 谱隙。

3. Adjoint dynamics. For $u_t + Lu = 0$, $u(0) = g$, and $v_t - L^*v = 0$, $v(T) = h$, both with zero Dirichlet data, prove $(v(0), g) = (u(T), h)$.

Solution. The formal adjoint is defined by $(Lu, v) = (u, L^*v)$ on the zero-boundary smooth functions, including the coefficient derivative in the adjoint drift. Therefore $\frac{d}{dt}(u, v) = -(Lu, v) + (u, L^*v) = 0$. Evaluate at $0, T$. The backward adjoint problem is prescribed terminal data; it is not a forward heat problem with the diffusion sign reversed arbitrarily. □

4. Galerkin approximation for Poisson's equation. For nested spans of a complete V-orthogonal basis, let $\int Du_m \cdot Dw_k = \int f w_k$ for $k \le m$. Show weak convergence to the zero-Dirichlet Poisson solution.

Solution. Testing with u_m and Poincaré bound $\|u_m\|_V$ uniformly. Each weak subsequential limit satisfies the equation on every fixed span, hence on V by density. Uniqueness identifies every subsequence. In fact one has

stronger convergence: if u is the weak solution, $\int D(u-u_m)\cdot Dv_m = 0$, so

$$\|D(u-u_m)\|_2 \le \inf_{v_m\in V_m} \|D(u-v_m)\|_2 \longrightarrow 0.$$

The finite systems are invertible since their gradient Gram matrices are positive definite. No L^2 orthonormality of this exercise's basis is needed. □

5. Identifying a weak time derivative. If $u_k \rightharpoonup u$ in L^2V and $u_k' \rightharpoonup z$ in L^2V', show $z = u'$.

Solution. For $\phi \in C_c^1(0,T)$, $w \in V$, $\int \langle u_k', w\rangle \phi = -\int (u_k, w)\phi'$. Both sides converge by the stipulated weak convergences. The resulting identity is exactly the definition of the V'-valued distributional derivative. □

6. The essential supremum survives a weak limit. If $u_k \rightharpoonup u$ in $L^2(0,T;H)$ and $\|u_k(t)\|_H \le C$ a.e., show $\|u(t)\|_H \le C$ a.e.

Solution. For any measurable $E \subset (0,T)$, $\mathbf{1}_E u \in L^2(0,T;H)$ is a valid fixed weak test. Thus

$$\int_E \|u\|_H^2 = \lim_k \int_E (u_k, u) \le C \int_E \|u\|_H.$$

Choosing $E = \{\|u\|_H > C + \delta\}$ makes its integrand difference at least $\delta\|u\|_H$, so E has measure zero. Take $\delta = 1/j$. This argument works even without assuming H separable. □

一致上界与弱极限

逐时刻弱收敛并非已知。用固定测试函数 $\mathbf{1}_E u$ 把问题化为每个时间集合上的积分不等式，直接得到极限的本质上确界。

7. Pointwise decay from a positive potential. For $u_t - \Delta u + cu = 0$, zero Dirichlet data and $c \ge \gamma > 0$, prove $|u(x,t)| \le Ce^{-\gamma t}$.

Solution. Set $z = e^{\gamma t}u$. Then $z_t - \Delta z + (c-\gamma)z = 0$ and the potential is nonnegative. Comparison with the constants $\pm\|g\|_\infty$ yields $|z| \le \|g\|_\infty$ on every finite cylinder. The constant is independent of its final time, proving the assertion. □

8. Positivity with a bounded signed potential. For the same equation, $g \ge 0$ and bounded c, show $u \ge 0$.

Solution. Choose $\lambda \ge \max(0, -\inf c)$ and $z = e^{-\lambda t}u$. Then $z_t - \Delta z + (c+\lambda)z = 0$. The minimum principle gives $z \ge 0$, and multiplication by $e^{\lambda t} > 0$ proves the claim. □

9. The second-order coercive estimate, including its boundary terms. Prove the source inequalities (54) in §7.1.3 and (59) in §7.2.3:

$$\beta\|u\|_{H^2}^2 \le (Lu, -\Delta u) + \gamma\|u\|_2^2 \qquad (u \in H^2(U) \cap V). \tag{7.15}$$

Solution. Take first smooth $u = 0$ on ∂U and smooth symmetric A. Expand the divergence operator as $Lu = -a^{ij}u_{ij} + \tilde{b}^i u_i + cu$. For the principal term put $I = \int_U a^{ij}u_{ij}u_{kk}$. Integrate first in k and then in j:

$$\begin{aligned} I = &\int_U a^{ij}u_{ik}u_{jk} + \int_U \big((\partial_j a^{ij})u_{ik}u_k - (\partial_k a^{ij})u_{ij}u_k\big) \\ &+ \int_{\partial U} \big((A : D^2u)\,\partial_\nu u - (D^2u\,A\nu)\cdot Du\big). \end{aligned}$$

The volume errors are bounded by $\delta\|D^2u\|_2^2 + C_\delta\|Du\|_2^2$. The first volume term is at least $\theta\|D^2u\|_2^2$. The boundary expression must not be discarded.

Write $q = \partial_\nu u$, $\Pi = I - \nu\otimes\nu$, and $a_\tau = \Pi A\nu$. Since the tangential derivatives of u vanish, $Du = q\nu$, $\Pi D^2u\,\nu = \nabla_\tau q$, and the tangential Hessian is $q\mathscr{S}$, where $\mathscr{S} = D_\tau\nu$ is the shape operator with this sign convention. Cancellation of the pure normal second derivative leaves precisely

$$(A : D^2u)q - q\nu\cdot D^2uA\nu = q^2\,\mathrm{tr}(\Pi A\Pi\mathscr{S}) + q\,a_\tau\cdot\nabla_\tau q.$$

Tangential integration by parts on the closed boundary turns its integral into

$$\int_{\partial U} q^2\left(\mathrm{tr}(\Pi A\Pi\mathscr{S}) - \tfrac{1}{2}\,\mathrm{div}_\tau\, a_\tau\right).$$

The coefficient is bounded by the C^1 norm of A and boundary curvature. A tubular-neighborhood trace estimate gives, for any $\delta > 0$,

$$\|Du\|_{L^2(\partial U)}^2 \le \delta\|D^2u\|_2^2 + C_\delta\|Du\|_2^2.$$

For completeness this follows by writing the boundary value of $|Du|^2$ as its value at an interior normal offset plus the normal integral of its derivative, averaging offsets, and applying Young to $|Du||D^2u|$; a finite collar cover has bounded Jacobians. Thus the boundary term costs only the same small highest-order norm and a lower-order norm. The drift and potential paired with $-\Delta u$ cost $\delta\|D^2u\|_2^2 + C_\delta\|u\|_{H^1}^2$. It follows that

$$(Lu, -\Delta u) \ge c_1\|D^2u\|_2^2 - C_1\|u\|_{H^1}^2.$$

Finally $\|Du\|_2^2 = -(u, \Delta u)$, so $\|Du\|_2^2 \le \delta\|D^2u\|_2^2 + C_\delta\|u\|_2^2$. Absorb this term and add a sufficiently large multiple of $\|u\|_2^2$ to get (7.15). Smooth zero-trace functions are dense in $H^2\cap V$ on a smooth domain; all final volume pairings are continuous in H^2, extending the estimate. Both source inequalities are the same estimate. Our earlier reconstruction proofs avoid needing it inside the Galerkin regularity passage, but it remains a useful independent result. □

10. Telegraph equation uniqueness. For $u_{tt} + du_t - u_{xx} = f$ on $(0,1)$, with zero endpoint values and two prescribed initial data, prove uniqueness for any real constant d.

Solution. For a difference w let $E = \frac{1}{2}\int_0^1 (w_t^2 + w_x^2)$. The endpoint term vanishes since $w_t = 0$ there. Then $E' = -d\int w_t^2 \le 2|d|E$. Zero initial energy and Gronwall imply $E = 0$; zero endpoint values give $w = 0$. For $d < 0$ energy can grow, but uniqueness still holds. □

11. Clamped beam uniqueness. For $u_{tt} + u_{xxxx} = 0$, $u = u_x = 0$ at $0, 1$, prove uniqueness.

Solution. For a difference let $E = \frac{1}{2}\int_0^1 (w_t^2 + w_{xx}^2)$. Twice integrating gives $\int w_{xxxx} w_t = \int w_{xx} w_{txx}$: the boundary terms are $[w_{xxx} w_t - w_{xx} w_{tx}]_0^1 = 0$ since both clamped data are constant in time. Thus $E' = 0$, $E(0) = 0$, and $w_{xx} = 0$. An affine function with zero clamped data is zero. □

12. Resolvent identities. Establish the source identities (12)–(13) in §7.4.1.

Solution. On $D(G)$ the difference of the two operators is $(\mu I - G) - (\lambda I - G) = (\mu - \lambda)I$. Multiplication on the left by R_λ and on the right by R_μ gives $R_\lambda - R_\mu = (\mu - \lambda) R_\lambda R_\mu$. Interchange parameters to get commutation when they differ; equality of parameters is automatic. The intermediate vectors are in $D(G)$, so no multiplication of an undefined unbounded expression occurs. □

13. Passing a closed generator through an integral. For $x \in D(G)$ justify $G\int_0^\infty e^{-\lambda t} S(t)x\,dt = \int_0^\infty e^{-\lambda t} GS(t)x\,dt$, $\lambda > 0$.

Solution. Equip $D(G)$ with $\|y\|_G = \|y\| + \|Gy\|$; closedness makes this a Banach space. The integrand is graph-norm continuous because $GS(t)x = S(t)Gx$, and its graph norm is at most $e^{-\lambda t}(\|x\| + \|Gx\|)$. Riemann sums on finite intervals therefore converge in graph norm, and the tails converge there as well. Applying the continuous map $G : (D(G), \|\cdot\|_G) \to X$ proves the identity. □

零边界不消去全部边界项

$u = 0$ 只消去切向一阶导数，法向导数 q 可不为零。纯法向二阶项抵消后，仍有曲率项和 $q\nabla_\tau q$；须切向分部积分并用带小常数的迹估计吸收。

14. Heat convolution on two different spaces. Let $S(t)g = \Phi_t * g$, $S(0) = I$. Show it is a contraction semigroup on $L^2(\mathbb{R}^n)$, but not a strongly continuous semigroup on $L^\infty(\mathbb{R}^n)$.

Solution. The Gaussian is nonnegative of mass one, so Young's inequality gives $\|S(t)\|_{2\to 2} \le 1$. Gaussian convolution gives $S(t+s) = S(t)S(s)$. On C_c^∞, approximate identities give $S(t)g \to g$ in L^2; density and contraction extend this to all L^2. On L^∞ the same convolution still contracts the norm. The failed axiom is strong continuity: take $g = \mathbf{1}_{\{x_1>0\}}$. Its smoothed value tends to $1/2$ as $x_1 \downarrow 0$, so for every $t > 0$, $\|S(t)g - g\|_\infty = 1/2$ (in the essential-supremum sense). The source's phrase "not a contraction semigroup" refers to this continuity failure, not to failure of the norm inequality. □

图范数使积分合法

只知道被积函数在 X 中可积不足以交换无界算子。同时控制函数及其 G 像，等于在完备图范数空间中积分。

15. Invariance of powers of the generator domain. Define $D(G^k) = \{x \in D(G^{k-1}) : G^{k-1}x \in D(G)\}$. Show $S(t)D(G^k) \subset D(G^k)$.

Solution. For $k = 1$ this is Theorem 7.17. Inductively, for $x \in D(G^k)$, one has $S(t)x \in D(G^{k-1})$ and $G^{k-1}S(t)x = S(t)G^{k-1}x \in D(G)$ by the $k = 1$ case. This proves invariance and also $G^k S(t)x = S(t)G^k x$. □

同一公式依赖所选空间

热卷积在 L^∞ 仍满足 $\|S(t)\| \le 1$，但阶跃函数阻止强连续。不能仅核对卷积律与范数，就宣布得到 C_0 半群。

16. Smooth spatial heat orbits from compactly supported data. For the Dirichlet heat semigroup and $g \in C_c^\infty(U)$, show $S(t)g \in C^\infty(U)$ for every $t \ge 0$.

Solution. Here $G = \Delta$, $D(G) = H^2 \cap V$. Every $\Delta^j g$ is again compactly supported smooth, so $g \in \bigcap_{k\ge 1} D(G^k)$. Problem 15 preserves these domains. Repeated elliptic regularity gives $D(G^k) \subset H^{2k}(U)$: apply the elliptic estimate to $Gu \in D(G^{k-1})$ inductively. For any spatial derivative order r, choose $2k > r + n/2$ and use Sobolev embedding. This proves the claim, including $t = 0$. Abstract domain invariance alone would not prove smoothing of arbitrary L^2 data; that stronger heat property needs additional estimates. □

7.6 References and Proof Dependencies

The primary mathematical source is Lawrence C. Evans, *Partial Differential Equations*, second edition, Chapter 7, printed pp. 349–423 (local English PDF pp. 366–440). The Chinese scan corresponds to printed pp. 248–299 (PDF pp. 266–317). Selected comparisons and the source theorem map are recorded in `docs/CHAPTER-07.md`. All English exposition, proof reconstructions, solutions, annotations and diagrams in this chapter are editorial writing.

Evans points to Ladyzhenskaya, J.-L. Lions, Lions–Magenes and Wloka for evolution equations; Davies for Harnack's calculation; Krylov, Ladyzhenskaya–Solonnikov–Uraltseva and Lieberman for further regularity; John and Treves for the Fourier approach; Friedman and Yosida for semigroups. Problem 9 is linked to Brezis–Evans, *Archive for Rational Mechanics and Analysis* 71 (1979), 1–13, and Ladyzhenskaya–Uraltseva. These source bibliography pointers were not independently consulted.

What is proved and what is assumed. The Galerkin constructions, initial-trace passage, integrated-test wave uniqueness, regularity inductions, smooth principal-part Harnack inequality, finite propagation under a smooth optical-distance hypothesis, vanishing viscosity construction, Fourier multiplier bound, generator properties, contraction Hille–Yosida theorem, and both PDE applications have expanded proofs. All 16 source exercises have editorial solutions; Problem 9 retains and estimates the boundary term explicitly. Hilbert weak compactness, Banach contraction, the closed graph theorem, Bochner integration, Fourier Plancherel and finite-dimensional Schur triangularization are prerequisites. Sobolev approximation, traces, extension and embedding come from Chapter 5; the Dirichlet spectral basis, Lax–Milgram and elliptic boundary estimates come from Chapter 6. The local flow-box construction uses classical smooth ODE theory.

Scope boundaries. General rough-coefficient Harnack theory, global smooth optical distances, nonsymmetric L^2 well-posedness under real spectrum alone, and analytic semigroup smoothing are not proved. Our strong maximum proof additionally assumes time-independent principal coefficients. The Fourier uniqueness class is specified explicitly; autonomous semigroups require time-independent coefficients. This chapter has not received independent mathematical review.